

WASP-2b

R-1058

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_151021 · created 2026-07-30T11:20:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457316.6922 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.047
Transit depth (Rp/Rs)^2	0.74 +/- 0.81 [%]
Orbital Inclination (inc)	80.98 +/- 2.92
Airmass coefficient 1 (a1)	1.66 +/- 0.85
Airmass coefficient 2 (a2)	-0.34 +/- 0.14
Scatter in the residuals of the lightcurve fit is	48.29 %
Best Comparison Star	None
Optimal Aperture	4.32
Optimal Annulus	9.37
Transit Duration (day)	0.018 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.086 ± 0.047 vs 0.1326 (deviation 0.97σ)
Duration fit vs published	0.018 d vs 0.0737 d (deviation 1.89σ)
Mid-time O–C	-8.8 min
Depth SNR	1.8
Residual scatter	48.29 %

Light curve

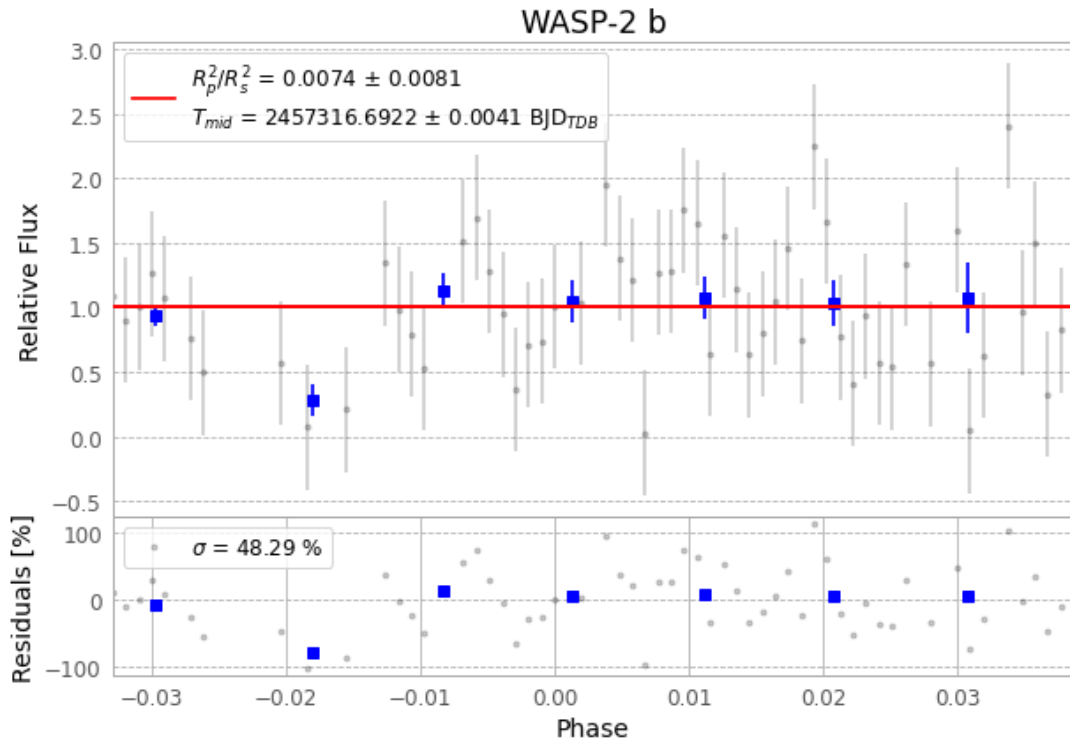


Figure 1 — FinalLightCurve_WASP-2 b_2015-10-20.png (SHA-256 700a56f232ec3152...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [200, 177], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0248fe2ebef92c61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2151021025113.FITS → WASP-2151021063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-151021.log (065f3274565c...), FinalLightCurve_WASP-2 b_2015-10-20.png (700a56f232ec...), FinalLightCurve_WASP-2 b_2015-10-20.csv (b14e8b4a1c23...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:20Z.